Lab 1: Requirements Table - Self-Service Coffee Kiosk

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Scenario: Self-Service Coffee Kiosk in a busy cafe: customers choose one of 3 coffee types (Espresso, Americano, Latte) in 2 sizes (Small, Large), pay by credit card only, and receive a printed receipt. The kiosk uses a touch screen, a receipt printer, and stores menu and pricing data.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Review Comments (revision made)
FR-001	Functional	The system shall allow a customer to select a coffee type (Espresso, Americano or Latte) from the touch-screen menu.	High	When a customer taps "Espresso", the item is highlighted and the screen moves to size selection within 1 s. Repeated for all 3 types.	Core ordering functionality.	Given in handout. Listed the 3 menu items so each one is tested.
FR-002	Functional	The system shall allow the customer to choose a drink size (Small or Large) for the selected coffee and shall show the price for that size.	High	For all 6 type/size combinations, the price shown on the summary screen equals the price stored in the menu table.	Price depends on size; the customer must see the correct amount before paying.	"Show price" was vague; now checked against the menu table.
FR-003	Functional	The system shall accept payment by credit card through the payment gateway and shall confirm the order only after the gateway approves the payment.	High	Gateway sandbox test: an approved test card sets the order to PAID and shows a confirmation; a declined test card leaves the order unpaid and shows "Payment declined".	Credit card is the only payment method; prevents handing out unpaid drinks.	Added the declined-card case so the test has a clear pass/fail.
FR-004	Functional	The system shall print a receipt showing order number, coffee type, size, price, date/time and the last 4 digits of the card after a successful payment.	High	After a paid test order the receipt contains all 6 fields, and printing starts within 5 s of payment approval.	Proof of purchase; the order number is used to collect the drink.	Listed the exact receipt fields instead of "order details".
FR-005	Functional	The system shall allow an authenticated administrator to add, edit or disable menu items and their prices.	Medium	Admin changes Small Americano from Rs 120 to Rs 130 and saves; the next customer session shows Rs 130. A wrong admin PIN is rejected.	Prices change; the cafe manager must update them without a developer.	Added the authentication check so customers cannot change prices.
NFR-001	Nonfunctional	The system shall complete any order (selection to payment) in under 60 s.	High	Kiosk logs for 30 timed test orders show the time from first tap to payment approval is below 60 s for every order.	Ensures quick service during peak hours.	Given in handout. Fixed the sample size (30 orders) so "consistently" is measurable.
NFR-002	Nonfunctional	The system shall never store the full card number or CVV on the kiosk, and shall send card data to the payment gateway only over TLS 1.2 or higher.	High	After 10 test payments, the kiosk database, logs and receipts contain no full card number or CVV (last 4 digits only), and a network capture shows only TLS 1.2+ connections to the gateway.	Protects customers and keeps the kiosk PCI-DSS compliant.	Split "secure payments" into two checks that can be inspected.

Actors

Actor	Kind	Goal
Customer	Primary (person)	Orders and pays for coffee at the kiosk.

Actor	Kind	Goal
Administrator (Cafe Manager)	Primary (person)	Maintains menu items and prices.
Payment Gateway	Secondary (external system)	Authorizes credit card payments.
Receipt Printer	Secondary (device)	Prints the customer receipt.

Use Cases

ID	Use Case	Actors	Traces to
UC-01	Select Coffee Type	Customer	FR-001
UC-02	Choose Drink Size	Customer	FR-002
UC-03	Place Order	Customer	FR-001..FR-004, NFR-001
UC-04	Pay by Credit Card	Customer, Payment Gateway	FR-003, NFR-002
UC-05	Print Receipt	Receipt Printer	FR-004
UC-06	Manage Menu & Prices	Administrator	FR-005
UC-07	Retry Payment	Customer, Payment Gateway	FR-003 (declined card)

Relationships: UC-03 Place Order «include» UC-04 Pay by Credit Card and «include» UC-05 Print Receipt (always performed). UC-07 Retry Payment «extend» UC-04 (only when the card is declined).